Unit no. ° 3

LINES IN THE PLANE

1-16 October 2026

I Direction vector and Cartesian equation

Definition

A direction vector of a line is a vector **non-zero** with the direction of this line. If $A \neq B$, AB directs (AB). Every line has an equation $ax + by + c = 0$ with $(a, b) \neq (0, 0)$. A direction vector is $\begin{pmatrix} -b \\ a \end{pmatrix}$.

Example 3.1

For $A(-2; 4)$ and $B(6; 2)$, calculate $\overrightarrow{AB}$ then verify that $\vec{u}(4; -1)$ directs (AB).

Solution:

Establish an equation using the determinant

The line through $A(x_A; y_A)$ and directed by $\vec{u}(\alpha; \beta) \neq 0$ has equation

$$\beta(x - x_A) - \alpha(y - y_A) = 0.$$

Write a proof.

Solution:

Watch out. The zero vector does not direct any line. The sign of the determinant depends on the order of the vectors, but whether it is zero does not depend on them.

II Determine an equation of a line

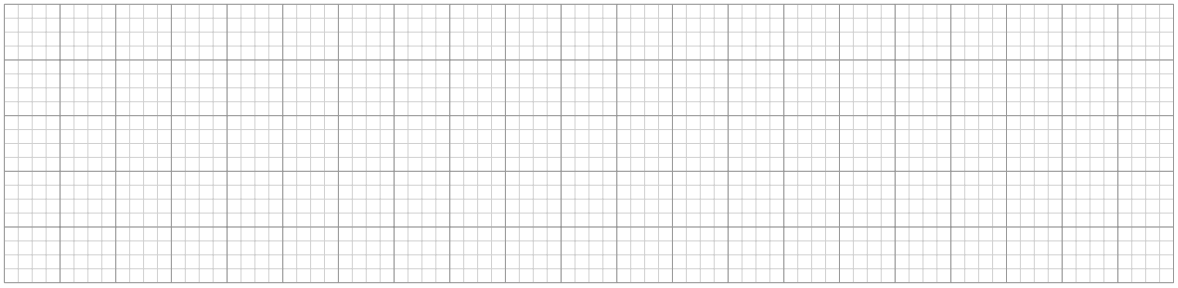
Method

From a point and a direction vector: choose a, b then replace x, y by the coordinates of the point to find c. From two distinct points: first calculate a direction vector.

Example 3.2

Determine a Cartesian equation of the line passing through $A(2; -1)$ and directed by $\vec{u}(3; 2)$.

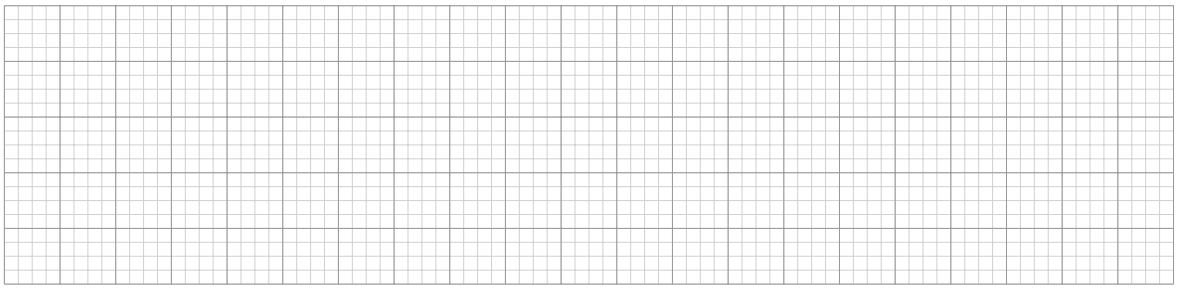
Solution:



Example 3.3

Determine a Cartesian equation of (AB) for $A(3; 3)$ and $B(2; -2)$.

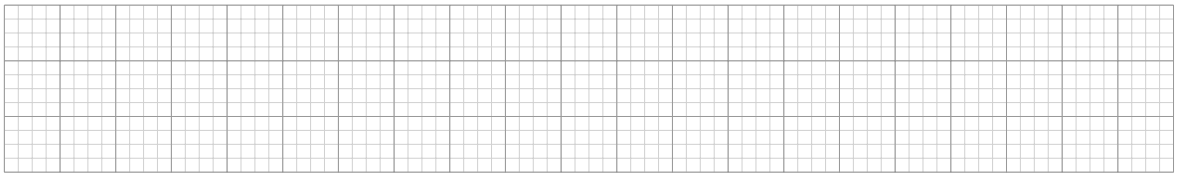
Solution:



Example 3.4

Is point $P(1; 2)$ lie on $d : 2x - 3y + 4 = 0$? Does the point $Q(2; 1)$ lie on it?

Solution:



III Slope-intercept equation, slope and affine function

Definitions

A non-vertical line has a unique equation of the form $y = mx + p$. m is the slope (gradient), p the y-intercept. It represents the affine function $x \mapsto mx + p$; $(1; m)$ is a direction vector. A vertical line has an equation $x = k$; it is not the graph of a function of x. A horizontal line has an equation $y = k$.

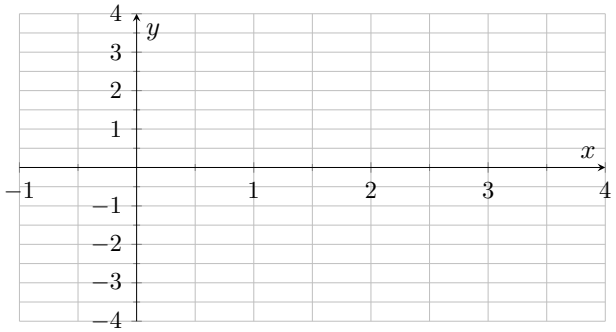
Slope calculation

If $x_A \neq x_B$, the slope of (AB) is $m = \dfrac{y_B - y_A}{x_B - x_A}$. To find p, then we use $y_A = mx_A + p$.

Example 3.5

Determine the equation y = mx + p of 2 $x + 3y - 1 = 0$, then draw $d : y = -2x + 3$.

Solution:



Example 3.6

Find the affine function f such that $f(0) = 3$ and $f(-1) = 5$, then the affine function g whose graph passes via $C(-1; 2)$ and $E(5; -2)$.

Solution:

IV Relative positions of two lines

Properties

Two lines are intersecting, strictly parallel or coincident. For $y = mx + p$ and $y = m'x + p'$:

· if $m \neq m'$, they intersect;

· if $m = m'$ & $p \neq p'$, they are strictly parallel;

· if $m = m'$ & $p = p'$, they coincide.

For Cartesian equations, we compare directions using the determinant. If the directions are collinear, we then test whether a point belongs to both to distinguish coincident lines from strictly parallel lines.

Example 3.7

Determine an equation of d through $A(2; 3)$ and parallel to $d' : x - 3y - 3 = 0$.

Solution:

Example 3.8

Study the relative positions:

1. $2x - y + 1 = 0$ and $4x - 2y + 2 = 0$;
2. $x = 2$ and $x = -1$;
3. $y = 2x - 3$ and $y = -x + 3$.

Solution:

V Intersection: solving by substitution

Definition

Solving a system of two equations in x, y, means finding all pairs $(x; y)$ satisfying the two equations simultaneously. For two equations of lines: an intersection gives a solution pair; strictly parallel lines give no solution; coincident lines give infinitely many solutions.

Substitution

Make one unknown the subject of one equation, substitute its expression into the other, solve, then find the second unknown. Check the pair in both original equations.

Example 3.9

Solve $\begin{cases} -2x + y + 3 = 0, \\ 3x - 5y - 2 = 0. \end{cases}$

Solution:

Example 3.10

Solve by substitution:

(a) $\begin{cases} -2x + y + 3 = 0 \\ x - 4y + 5 = 0 \end{cases}$ (b) $\begin{cases} 2x + 3y - 10 = 0 \\ 4x + 5y - 16 = 0. \end{cases}$

Solution:

VI Linear combination and Python

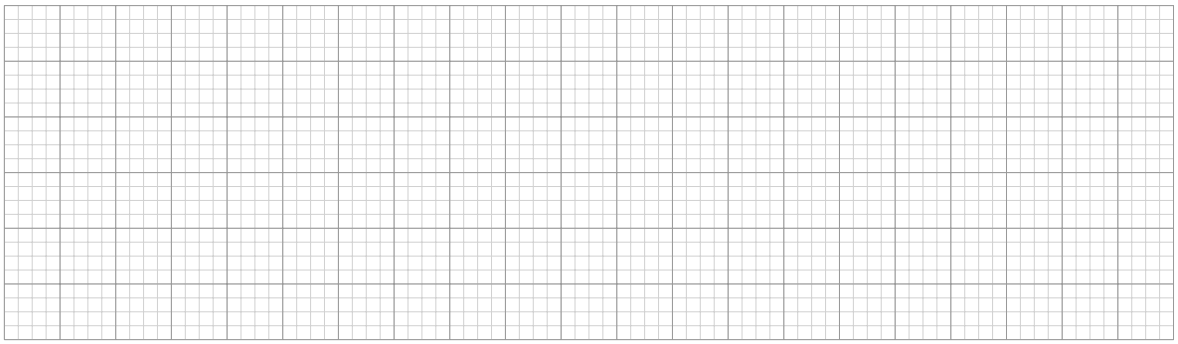
Linear combination

Multiply the equations by non-zero numbers to obtain opposite coefficients of one unknown; add them to eliminate this unknown. Keep one equation of the system to calculate the other unknown.

Example 3.11

Solve $\begin{cases} 2x+3y+1=0 \\ 3x-4y-2=0. \end{cases}$ Then redo the two systems from the previous example by elimination.

Solution:



```
def determinant(u, v):
    return u[0]*v[1] - u[1]*v[0]

def alignes(A, B, C):
    AB = (B[0]-A[0], B[1]-A[1])
    AC = (C[0]-A[0], C[1]-A[1])
    return determinant(AB, AC) == 0
```

Example 3.12

Predict `alignes((0,0), (2,1), (6,3))` then `alignes((0,0), (2,1), (6,4))`.

Solution:

